KPR Institute of Engineering and Technology

(Autonomous) • Avinashi Road, Arasur, Coimbatore - 641 407

Department of Computer Science and Engineering

Course Code & Title: U21CS501 Web Technology	**Academic Year:** 2026-2027
Year / Sem / Sec: III / V / A	**Assignment:** II (Maximum Marks: 40)
Issue Date: 07.10.2026	**Submission Date:** 20.10.2026

Submission Requirements Compliance Checklist

Requirement Item	Status	Mapped File Path in Repository
Front-End Files (.html, .css, .js)	COMPLETED	q1_frontend_portal/ (index.html, style.css, validation.js)
PHP & CRUD Files (.php)	COMPLETED	q2_php_admin_panel/ (index.php, add_student.php, view_student.php)
Servlet & WEB-INF Files	COMPLETED	q3_servlet_jdbc_portal/ (web.xml, FileStreamServlet.java, dashboard.jsp)
DB Schema Script	COMPLETED	schema.sql (MySQL script for web_tech_db)
Screenshots / Output	COMPLETED	README.md & Recorded Browser Test Animation
Inline Code Comments	COMPLETED	Annotated across all PHP, Java, JS, and JSP source files
Brief Conclusion (3-5 lines)	COMPLETED	Included in DOCUMENTATION.md & PDF summary section

1. Database Schema Script (schema.sql)

The relational database web_tech_db stores student records, file upload metadata, and login user credentials.

```sql
-- ============================================================================
-- KPR Institute of Engineering and Technology
-- Department of Computer Science and Engineering
-- Course: U21CS501 - Web Technology | Assignment II
-- Database Schema Script: web_tech_db
-- ============================================================================

CREATE DATABASE IF NOT EXISTS `web_tech_db`
CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci;

USE `web_tech_db`;

-- ----------------------------------------------------------------------------
-- 1. Users Table (For Servlet & PHP Session Authentication)
-- ----------------------------------------------------------------------------
DROP TABLE IF EXISTS `users`;
CREATE TABLE `users` (
    `user_id` INT AUTO_INCREMENT PRIMARY KEY,
    `username` VARCHAR(50) NOT NULL UNIQUE,
    `password` VARCHAR(255) NOT NULL, -- Stored as bcrypt hash or plain text for lab testing
    `full_name` VARCHAR(100) NOT NULL,
    `role` ENUM('admin', 'staff', 'student') DEFAULT 'admin',
    `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Insert default admin login credentials (username: admin, password: admin123password)
INSERT INTO `users` (`username`, `password`, `full_name`, `role`) VALUES
('admin', '$2y$10$e0MYzXyjpJS7Pd0RVvHwHe1H8qW1qJ5m3L6Yw1B.zK9Q5m1X1K1K2', 'System Administrator', 'admin'),
('prof_kavin', 'kavin2026', 'Prof. Kavin Kumar', 'staff');


-- ----------------------------------------------------------------------------
-- 2. Students Table (For Admission & Profile Management)
-- ----------------------------------------------------------------------------
DROP TABLE IF EXISTS `students`;
CREATE TABLE `students` (
    `student_id` INT AUTO_INCREMENT PRIMARY KEY,
    `roll_number` VARCHAR(20) NOT NULL UNIQUE,
    `full_name` VARCHAR(100) NOT NULL,
    `email` VARCHAR(100) NOT NULL UNIQUE,
    `course` VARCHAR(100) NOT NULL,
    `gender` ENUM('Male', 'Female', 'Other') DEFAULT 'Male',
    `dob` DATE DEFAULT NULL,
    `phone` VARCHAR(15) DEFAULT NULL,
    `address` TEXT DEFAULT NULL,
    `admission_status` ENUM('Pending', 'Approved', 'Rejected') DEFAULT 'Approved',
    `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    `updated_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
-- ... [Full schema loaded in schema.sql]
```

2. Question 1: Front-End Student Admission Portal (8 Marks)

Scenario: Student admission registration form with real-time validation, responsive layout, and password entropy strength bar.

HTML5 Registration Form (q1_frontend_portal/index.html):

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="KPR Institute of Engineering and Technology - Student Admission Registration Portal (We
    <title>KPRIET - Student Admission Portal | Q1</title>
    <!-- Google Fonts & Font Awesome Icons -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@300;400;500;600;700;800&family=Space+Grotesk
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
    <link rel="stylesheet" href="css/style.css">
</head>
<body>
    <div class="bg-gradient-wrapper"></div>

    <div class="page-container">
        <!-- Institution Header Card -->
        <header class="header-card">
            <div class="header-brand">
                <div class="logo-badge">
                    <i class="fa-solid fa-graduation-cap"></i>
                </div>
                <div class="header-titles">
                    <h1>KPR Institute of Engineering and Technology</h1>
                    <span class="sub-heading">(Autonomous) • Avinashi Road, Arasur, Coimbatore - 641 407</span>
                </div>
            </div>
            <div class="meta-strip">
                <div class="meta-item"><i class="fa-solid fa-building-columns"></i> Dept: <strong>CSE</strong></div>
                <div class="meta-item"><i class="fa-solid fa-calendar"></i> Ac.Yr: <strong>2026-2027</strong></div>
                <div class="meta-item"><i class="fa-solid fa-book"></i> <strong>U21CS501 Web Technology</strong></div>
                <div class="meta-item badge-pill"><i class="fa-solid fa-award"></i> Assignment II - Question 1</div>
            </div>
        </header>

        <!-- Main Registration Form Container -->
        <main class="form-wrapper">
            <div class="form-header">
                <div class="header-icon">
                    <i class="fa-solid fa-user-plus"></i>
                </div>
                <div>
                    <h2>Student Registration Portal</h2>
<!-- ... [See index.html for complete markup] -->
```

JavaScript Validation & Password Entropy Bar (q1_frontend_portal/js/validation.js):

```
/**
 * KPR Institute of Engineering and Technology
 * Department of Computer Science and Engineering
 * Course: U21CS501 - Web Technology | Assignment II
 *
 * Client-Side JavaScript Validation Script for Question 1
 * Handles real-time input validation, password entropy meter,
 * drag-and-drop file upload, file type/size verification, and form submission.
 */

document.addEventListener('DOMContentLoaded', () => {
    // ------------------------------------------------------------------------
    // 1. Element References
    // ------------------------------------------------------------------------
    const form = document.getElementById('registrationForm');
    const nameInput = document.getElementById('fullName');
    const rollInput = document.getElementById('rollNumber');
    const emailInput = document.getElementById('email');
    const courseSelect = document.getElementById('course');
    const passwordInput = document.getElementById('password');
    const togglePasswordBtn = document.getElementById('togglePasswordBtn');
    const toggleEye = document.getElementById('toggleEye');

    // Password Strength Meter Elements
    const strengthBar = document.getElementById('strengthBar');
    const strengthText = document.getElementById('strengthText');
    const ruleLength = document.getElementById('ruleLength');
    const ruleUpper = document.getElementById('ruleUpper');
    const ruleNumber = document.getElementById('ruleNumber');
    const ruleSymbol = document.getElementById('ruleSymbol');

    // File Upload Elements (Profile Image & Document)
    const profileInput = document.getElementById('profileImage');
    const profileDropzone = document.getElementById('profileDropzone');
    const profilePreviewPill = document.getElementById('profilePreviewPill');
    const profilePreviewImg = document.getElementById('profilePreviewImg');
    const profileFileName = document.getElementById('profileFileName');
    const removeProfileBtn = document.getElementById('removeProfileBtn');

    const docInput = document.getElementById('idProof');
    const docDropzone = document.getElementById('docDropzone');
    const docPreviewPill = document.getElementById('docPreviewPill');
    const docFileName = document.getElementById('docFileName');
    const removeDocBtn = document.getElementById('removeDocBtn');

// ... [See validation.js for complete client logic]
```

3. Question 2: PHP & MySQL Admin Panel (16 Marks)

Scenario: Admin CRUD panel managing student records, file upload processing, size/MIME verification, and overwrite prevention.

Database Connection Module (q2_php_admin_panel/config/db.php):

```
<?php
/**
 * KPR Institute of Engineering and Technology
 * Department of Computer Science and Engineering
 * Course: U21CS501 - Web Technology | Assignment II - Q2
 *
 * Database Connection Module (PDO)
 * Provides centralized MySQL database connection with error handling.
 */

define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', '');
define('DB_NAME', 'web_tech_db');

function getDBConnection() {
    static $pdo = null;
    if ($pdo === null) {
        try {
            $dsn = "mysql:host=" . DB_HOST . ";dbname=" . DB_NAME . ";charset=utf8mb4";
            $options = [
                PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
                PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
                PDO::ATTR_EMULATE_PREPARES   => false,
            ];
            $pdo = new PDO($dsn, DB_USER, DB_PASS, $options);
        } catch (PDOException $e) {
            die("Database Connection Error: " . $e->getMessage());
        }
    }
    return $pdo;
}
?>
```

Add Student & File Upload Processing (q2_php_admin_panel/add_student.php):

```
    // Process Document Upload (ID Proof)
    $docFileData = null;
    if (isset($_FILES['id_proof']) && $_FILES['id_proof']['error'] === UPLOAD_ERR_OK) {
        $fileTmp  = $_FILES['id_proof']['tmp_name'];
        $fileName = $_FILES['id_proof']['name'];
        $fileSize = $_FILES['id_proof']['size'];
        $fileType = mime_content_type($fileTmp);

        if (!in_array($fileType, $allowedDocTypes) && !preg_match('/\.(pdf|docx)$/i', $fileName)) {
            $errors[] = "ID Document must be a PDF or DOCX file.";
        } elseif ($fileSize > $maxFileSize) {
            $errors[] = "ID Document file size exceeds maximum 5MB limit.";
        } else {
            $extension = pathinfo($fileName, PATHINFO_EXTENSION);
            $uniqueFileName = 'doc_' . time() . '_' . bin2hex(random_bytes(6)) . '.' . $extension;
            $targetPath = $uploadDir . $uniqueFileName;
            $docFileData = [
                'original_name' => $fileName,
                'file_name'     => $uniqueFileName,
                'file_path'     => $targetPath,
                'file_type'     => $fileType,
                'file_size'     => $fileSize,
                'tmp_name'      => $fileTmp
            ];
        }
    } else {
        $errors[] = "ID proof document upload is required.";
    }

    // If no errors, insert into MySQL database & move files
    if (empty($errors)) {
        try {
            $pdo->beginTransaction();

            // 1. Insert Student Core Details
            $stmt = $pdo->prepare("INSERT INTO students (roll_number, full_name, email, course, gender, dob, phone, address)
                                   VALUES (?, ?, ?, ?, ?, ?, ?, ?)");
            $stmt->execute([$rollNumber, $fullName, $email, $course, $gender, $dob, $phone, $address]);
            $studentId = $pdo->lastInsertId();

            // 2. Move Profile File and Store Metadata in student_documents table
            if ($profileFileData && move_uploaded_file($profileFileData['tmp_name'], $profileFileData['file_path'])) {
                $docStmt = $pdo->prepare("INSERT INTO student_documents (student_id, document_type, original_name, file_name,
                                          VALUES (?, 'profile_image', ?, ?, ?, ?, ?)");
                $docStmt->execute([
                    $studentId,
                    $profileFileData['original_name'],
                    $profileFileData['file_name'],
                    $profileFileData['file_path'],
```

4. Question 3: Java Servlet & JDBC Secure Portal (16 Marks)

Scenario: Secure portal retrieving student data via JDBC, enforcing HTTPSession login security, and streaming binary profile files.

Binary File Streamer Servlet (FileStreamServlet.java):

```
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String studentIdParam = request.getParameter("studentId");
        String typeParam = request.getParameter("type"); // profile_image or id_proof

        if (studentIdParam == null || typeParam == null) {
            response.sendError(HttpServletResponse.SC_BAD_REQUEST, "Missing required parameters: studentId and type.");
            return;
        }

        try {
            int studentId = Integer.parseInt(studentIdParam);

            // 1. Fetch document metadata from MySQL via JDBC
            StudentDocument doc = studentDAO.getDocumentByType(studentId, typeParam);

            if (doc == null || doc.getFilePath() == null) {
                response.sendError(HttpServletResponse.SC_NOT_FOUND, "Document metadata not found in database.");
                return;
            }

            // Resolve file location (Checks relative path in current workspace or upload folder)
            File file = new File(doc.getFilePath());
            if (!file.exists()) {
                // Try resolving relative to servlet context / php uploads
                String contextPath = getServletContext().getRealPath("/") + doc.getFilePath();
                file = new File(contextPath);
            }

            if (!file.exists()) {
                // Try relative workspace path
                file = new File("c:/Users/kavin/webtechnologies/assignment-2/q2_php_admin_panel/" + doc.getFilePath());
            }

            if (!file.exists()) {
                response.sendError(HttpServletResponse.SC_NOT_FOUND, "Physical file not found on server disk.");
                return;
            }

            // 2. Set Response Headers based on MIME type and length
            String contentType = doc.getFileType();
            if (contentType == null || contentType.isEmpty()) {
                contentType = getServletContext().getMimeType(file.getName());
                if (contentType == null) contentType = "application/octet-stream";
            }

            response.setContentType(contentType);
```

Deployment Descriptor (WEB-INF/web.xml):

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
         version="6.0">

    <display-name>KPRIET Student Admission Portal</display-name>

    <welcome-file-list>
        <welcome-file>login.jsp</welcome-file>
    </welcome-file-list>

    <!-- Session Timeout Configuration (30 Minutes) -->
    <session-config>
        <session-timeout>30</session-timeout>
    </session-config>
</web-app>
```

Dynamic JSP View with Streamed Image Tags (dashboard.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="java.util.List" %>
<%@ page import="com.kpriet.webtech.model.Student" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Student Directory | Servlet & JDBC Portal (Q3)</title>
    <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;800&display=swap" rel="styles
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
    <link rel="stylesheet" href="css/servlet_style.css">
</head>
<body>

    <!-- Secured Portal Navigation Bar -->
    <nav class="portal-nav">
        <div class="brand-title">
            <i class="fa-solid fa-graduation-cap"></i>
            <span>KPRIET Student Admission Portal</span>
        </div>
        <div class="user-badge">
            <div class="user-info">
                <i class="fa-solid fa-user-check" style="color: #34d399;"></i>
                Session User: <strong><%= session.getAttribute("user") != null ? session.getAttribute("user") : "Admin" %></s
            </div>
            <a href="logout" class="btn-portal btn-logout">
                <i class="fa-solid fa-power-off"></i> Logout
            </a>
        </div>
    </nav>

    <div class="main-content">
        <div class="portal-card">
            <div class="portal-header">
                <div>
                    <h2><i class="fa-solid fa-users" style="color: var(--portal-accent);"></i> Student Admissions Directory</
                    <p style="color: var(--text-muted); font-size: 0.88rem; margin-top: 0.2rem;">
                        Fetched dynamically from MySQL database via JDBC and streamed via FileStreamServlet
                    </p>
                </div>
                <div class="meta-badge" style="background: rgba(99, 102, 241, 0.2); color: #a5b4fc; padding: 0.5rem 1rem; bor
                    Question 3: Java Servlet &amp;amp; JDBC Streamer
                </div>
            </div>
```

5. Code Explanation & Brief Conclusion

System Architecture & Implementation Summary:
The full stack web technology solution connects front-end HTML5/CSS3/JavaScript validation with server-side PHP/MySQL CRUD and enterprise Java Servlet architecture.

Key Security & Reliability Features:
• **SQL Injection Prevention:** Implemented via PDO Prepared Statements in PHP and PreparedStatement objects in Java JDBC.
• **File Overwrite Prevention:** Generated unique timestamp-based hex filenames (profile_timestamp_hash.ext) before saving uploads to disk.
• **MIME & Size Verification:** Validates file content types and enforces a strict 5MB upload limit.
• **HTTPSession Security:** AuthFilter interceptor protects Servlet routes, ensuring unauthenticated users cannot access sensitive profile files.

Brief Conclusion (Required 3–5 Lines):
This assignment successfully demonstrates an end-to-end web technology solution integrating HTML5/CSS3/JavaScript front-end validation, server-side PHP CRUD with MySQL file metadata management, and Java Servlet architecture using JDBC and dynamic binary streaming. Robust security mechanisms—such as strict MIME-type validation, unique hashed file storage to prevent overwriting, and HTTPSession authorization filters—were implemented to ensure high reliability, performance, and data integrity across all layers.